

Telepresence — the future of telephony

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Opportunities for future telepresence services, creating a sense of being present at a remote location, are outlined against a backdrop of telephony as an existing, highly successful telecommunications service. In providing a context for the papers of this themed issue, key telepresence technologies, the building blocks of future services, addressing issues in audio, data, video and virtual environments are reviewed. The practical complexity of commercial telepresence applications, where developments in component multimedia technologies must be complemented by advances in integration, interface and service, is also considered. The paper concludes with a commentary on early applications and a vision of more generic future telepresence solutions — ‘Immersive Conferencing’ and ‘Persistent Communities’.

1. The telephone heritage

Telepresence is enabling human interaction at a distance, creating a sense of being present at a remote location. As a well-established telepresence service, the telephone extends human speech and hearing. Whatever your metric, it is an outstandingly successful service — 600 million lines world-wide, 100% interoperability, comprehensive directory services, simple interface, consistent quality of service, low cost, and universal accessibility.

Moreover, the telephone has moulded both users and society to its capabilities and limitations. There is established telephone etiquette for both business and personal calls, and ‘user training’ is an assumed part of modern childhood. As we plot a course into a future of ubiquitous, multimedia telepresence we must be conscious of this heritage. The telephone is the ‘existence theorem’ for such universal services, but has also set a high level of customer and market expectations.

In the next millennium we are certain to be increasingly ‘telepresent’ both at work and at play, with an expanding market for telepresence services. As we move from a physical to an information economy, demand will grow for services which transport bits not atoms and which support the output of the human mind. Moreover, technology continues to advance at an unprecedented rate, and it can be suggested that we are reaching key breakpoints in client processing power and global network interconnectivity. Against a background of these favourable long-term trends, this paper overviews the status and prospects for telepresence, providing a context for the other themed papers.

We start with a review of key telepresence technologies, the building blocks of future services, addressing issues in

audio, data, video and virtual environments. This segmentation of media is a reflection of traditionally distinct technical disciplines, but we move on to consider the reality of telepresence as a complex multimedia service, where developments in component technologies must be complemented by advances in integration, interface and service. We conclude with a commentary on current vertical market applications and more generic future telepresence solutions.

2. Telepresence technologies

In setting the scene for papers on current and future work, the paper by Parke [1] also serves as enlightening historical background. His review charts the development of BT’s conferencing services from proprietary analogue systems in fixed studio locations, through gradual evolution and occasional revolution to standards-based digital services with widespread availability.

As we speculate on a potential explosion in telepresence services enabled by decreasing costs and increasing mobility in networks and consumer hardware, it is salutary to note that Parke chronicles several decades of remarkable technical advances but only a handful of commercial services. Indeed, pundits world-wide have been repeatedly over-optimistic in their predictions of market growth. Spectacular growth in PC ownership (180 million in 1995) and Internet hosts (40 million in 1996) confirms the potential for revolutionary developments, but in aspiring to truly mass-market telepresence services, it should be remembered that 50% of the world’s population has yet to make their first phone call. Achieving technical interoperability is certainly a prerequisite, but we will return later to the importance of parallel advances in aspects of

interface and service surround in pushing forward a market with a chequered history for matching technical capabilities to customer needs.

2.1 Audio

In real-time communications, audio is the core of the media experience. When receiving information it can be argued that 'a picture is worth a thousand words', but, lacking a direct visual output capability, human-to-human interaction is dominated by speech.

There are well-established and evolving standards for the delivery of 'telephone quality' monaural speech over ever-decreasing bandwidths. Gibson et al [2] discuss service advances which promise to make multiparty audio conferencing almost as accessible and routine as one-to-one telephony. Audio conferencing is currently a closely supervised, premium service, ideally suited to pre-arranged business meetings but inappropriate for informal or social discussions. Gibson describes developments in unattended conferencing which put the customer in control and offer flexible, cost-effective and scalable conferences.

However, monaural speech falls well short of our rich audio experience in the physical world, and bridging that gap would be a significant boost for telepresence. As coding techniques and processing power advance and network costs decline, we can anticipate higher audio bandwidths and multi-channel sound formats. Hollier et al [3] review the principles and promise of spatial audio, technology which can recreate a complete audio environment with directional sound sources and location-specific acoustics. The advantages of spatial audio have been proven in complex control environments such as aircraft cockpits and compelling three-dimensional sound for computer games is currently the focus of much commercial activity — benefits in more routine applications are readily envisaged.

Further opportunities are afforded by interpretative management of the audio information. A combination of speech processing and natural language techniques promises spoken language systems supporting direct man/machine dialogue. Once again, the pre-eminence of speech as the human 'output channel' suggests that this is certain to be a dominant element in future service interfaces. Westall et al [4] provide an introduction to this area.

Looking at wider technology trends, the PC, and computers in general, have stressed the importance of the visual display, and in application development, audio is typically a poor relation in the allocation of system resources such as CPU cycles. Practical issues such as audio interference in open-plan offices are one reason, but as we move to a future of interactive, mobile devices, with screen space at a premium, there will be increasing emphasis on

'auditory displays'. The creation and management of personal audio workspaces is a critical telepresence enabler, with miniaturised, wireless transducers competing with free-space solutions. Delivering flexible audio landscapes into hostile settings, such as open-plan and reverberant offices, presents substantial technical challenges, but with potential for commensurate rewards.

2.2 Data

Remote interaction and presentation of data is a key element of many telepresence services. Even asynchronous exchanges by e-mail or file transfer can be viewed as telepresence, extending our ability to communicate at a distance. Indeed, a tighter integration of real-time conferencing with established desktop groupware will be an important market development. Directories and interfaces will be required to support a seamless transition from asynchronous to real-time services, mirroring the reality of multi-faceted human communication around a single transaction or relationship. Even a one-off conference involves pre-meeting set-up and post-meeting 'wash-up' phases, which a customer would prefer to see integrated into a single service.

Within a real-time service, data-conferencing is typically used in parallel with other media for the direct human/human interaction. Russ [5] reviews progress in collaborative data sharing, and describes a trial audio-graphic conferencing service developed by BT. Shared access to files within a workgroup is already commonplace, and Russ points to a future of collaborative applications on a shared desktop.

In extending such services beyond the local team one aspect of data-conferencing requiring particular attention is security. For media such as audio and video, security equates primarily to the protection of privacy in real-time streams, whereas shared data may imply access to persistent file storage, typically protected within a corporate firewall. This issue is addressed by Cordell et al [6]. At one level, it is appropriate for us to address telepresence services in isolation from developments in underlying network protocols, trusting that gateways will smooth over problems of inter-working between a diversity of switched and packet-based networks. However, services must ultimately be 'network-aware', recognising and adapting to the data transport capabilities, and Cordell also considers wider aspects of Internet conferencing, including discussion of emerging standards.

Finally under the heading of data, we consider 'text chat', an alternative or supplement to speech, which is prevalent in current Internet games and chat rooms. Text dialogue, while tending to be regarded as a short-term expediency for overcoming limitations in network

performance, still has a long-term appeal where cost savings, persistence and pseudo-real-time nature are of direct benefit. For example, in applications that do not demand 100% attention, a text chat user can sustain more than one simultaneous session — a limited form of concurrent telepresence with multiple identities [7].

2.3 Video

In many respects, video is an obvious and desirable extension to audio in enhancing any telepresence experience. We have already referred to vision as the most powerful human ‘input channel’, and although sometimes subtle and overlooked, visual output in the form of gesture and body language is a key element of rich dialogue in a physical encounter. However, transmission and display of high-quality moving images is enormously costly compared with audio, and current options for intelligent, content-based processing and control are even more restricted than with speech signals.

Videoconferencing services have met with only limited commercial success, with image quality a frequent concern. It can be argued that television has established a benchmark for customer expectations, and reaction to the ‘pixelated jerk-vision’ achievable over traditional audio bandwidths is often disparaging. There are also many types of meeting where visual contact is relatively unimportant compared with audio and data interaction, and anecdotal evidence is peppered with references to minimised video windows and disconnected cameras in desktop applications. The greatest penetration of videoconferencing is in high-end ‘board room’ systems, which deliver acceptable image quality over high bandwidth links, and where discursive interaction is strongly supported by visual cues.

Uptake of multipoint videoconferencing services is similarly limited, with specific issues of scalability in the distribution and display of multiple images. ‘Continuous presence’, where all locations are tiled into a single image, is generally preferred to voice switching or manual control, but there is no sense in which participants achieve a feeling of co-presence in a shared meeting space. Lewis and Cosier [8] discuss the uptake of video-conferencing, focusing on social and interface challenges and seeking inspiration in the history and techniques of film and television.

Such service issues notwithstanding, there has been tremendous progress in technical standards and interoperability for video, and this is reviewed by Whybray et al [9]. Their paper also points towards future developments in content-based coding and processing which will open opportunities for more sophisticated yet cost-effective services.

Moving beyond general conferencing, video is an essential element of many specific telepresence services. In

remote surveillance, medical or engineering applications there may be objective requirements for image quality, and trade-offs can be made in meeting the particular needs. For example, a mixture of poor-quality moving and high-quality still images may be appropriate, or the quantifiable benefits may justify more expensive equipment and network connections. We will return to such ‘vertical market’ applications later.

This section concludes with an assertion that in the long-term video **will** be an assumed component of most telepresence services. Although we have outlined a number of factors which often make the cost-benefit equation unfavourable in the short term, some degree of benefit is invariably present, and will ultimately be realised in the face of declining costs.

2.4 Virtual environments

Our final telepresence technology is the all-encompassing area of virtual environments. At one level, virtual environments promise a faithful recreation of physical reality with complete sensory telepresence. Beyond that, they need not be constrained by physical laws and could even amplify or modify conventional human presence. More realistically in the immediate future, we are concerned with computer-generated three-dimensional graphical environments, which may include other media components such as real-time streamed audio or video. This flexible mixing of media — audio, video and data, real-time and persistent — in a single consistent environment, is the unique appeal of virtual environments.

In conferencing applications the focus is on multi-user or ‘inhabited’ virtual environments, which serve as shared spaces for communication and interaction. A key concept in such spaces is the ‘avatar’, a graphical embodiment through which the user interacts with their surroundings and other users. Depending on the technology and application, the avatar may be more or less realistic, with non-human forms common in multi-player games. Greenhalgh et al from Nottingham University [10] describe a relatively sophisticated virtual environment (‘Inhabiting the Web’) which was used for a series of meetings between BT Laboratories (BTL) and five UK universities over the SuperJanet network. Virtual environments can potentially deliver highly scalable telepresence services, and their paper discusses some of the performance issues and architectural options.

Of particular interest in non-business applications, avatars afford the option of anonymity, which provides equality between users and promotes open dialogue in a setting of safe intimacy. In exploring service concepts in this area, a recent publication describes ‘The Mirror’, a multi-user Internet virtual environment, based on the

emerging VRML standard [11]. The Mirror mixed professional content with social chat and interaction in a vision of 'Inhabited TV', which was delivered to two thousand viewers of the BBC series 'The Net' (see Fig 1).



Fig 1 Avatars 'inhabiting' a bouncy castle in The Mirror.

Although there is a wealth of options and opportunities in virtual environments for telepresence, it is clear from these and other trials that we remain in the early stages of experimentation. There are few standards for technical interoperability, and studies of interface and performance are in their infancy. Mortlock et al [12] report on the state of the art that can be achieved in a research setting. While virtual environments can portray complex behaviours, this is of limited value when user interaction is constrained by the mouse and keyboard, and their paper describes the use of gesture tracking to facilitate the transition from real to virtual world. Further advances in such intuitive, multi-modal interfaces will be important developments.

Traill et al [13] consider a complementary approach, which seeks to mix real and virtual worlds so as to maximise their respective strengths. They focus on collaborative working, and highlight a number of applications where 'enhanced reality' telepresence services offer more appropriate functionality than their fully virtual relations. As with video, virtual environments will find early niche applications in telepresence services which deliver benefit for a specific remote interaction as opposed to supporting general-purpose conferencing. Opportunities include remote manipulation in hostile locations and also control of miniature instruments, where virtual environments can deliver a translation of physical scale. Such applications may engage senses other than vision and hearing, with force feedback offering clear benefits.

In the face of so many options in applications and implementation — desktop versus room versus immersive, and the physical form of future computing 'appliances' are just two areas that we have not considered — our brief review can only hint at the issues and potential of virtual environments for telepresence. The 'dream' of immersive

telepresence is the driver for several long-term network and computing research programmes, such as the US Internet2 initiative [14], and many challenges remain. Not the least of these is identifying the most appropriate combination of technologies to serve specific market needs, and then packaging them into an attractive commercial service. Contrasting this technological complexity with our baseline telepresence service, the telephone, highlights the importance of **service** above **technology** and leads on to the next section of our discussions.

3. Telepresence services

In the preceding section we considered specific telepresence technologies, able to extend selected human senses and interactions at a distance. However, we are ultimately concerned with telepresence as a service, not as isolated technologies. Our interest is motivated by specific customer communications requirements, which can be satisfied by an appropriate combination of media and control.

The starting point for all communications is a customer need, a particular desire for contact. The nature of that need is then reconciled with the available technology in selecting an appropriate service. At present, the choice of technology for many people may be restricted to a phone call, fax, post or a physical meeting, with e-mail an increasingly prevalent option. However, we can anticipate a future of greatly increased diversity of media, where there will be a much wider choice in matching an appropriate degree of presence with that initial desire for contact.

Making that match, with a cost-effective and easy to use solution will be the key to commercial telepresence. Interoperability is perhaps the leading challenge for component technologies, and we have referred to the varying states of standards development in the areas of audio, data, video, and virtual environments. In this section we move on to consider the wider user experience, and to discuss specific service opportunities.

3.1 User interface

Returning once again to our baseline telepresence service, we speak of 'picking up the phone' to establish a telephony connection — a powerful expression of simplicity in a ubiquitous interface. However, even minor variations in national tones and dialling procedures can confuse the uninitiated, and greater complexity in mobile handsets, office PABX telephones and network services is fuelling an increasing weight of manuals and user guides. In multimedia communications we are far from 'picking up the camera' or even 'clicking the mouse' to initiate a dialogue. Day-one demand for world-wide services and global manufacturing ambitions will help to minimise international

disparities, but will do nothing to halt this spiral of complexity and ultimately customer confusion.

While this burgeoning complexity can be handled by the expert user in the short term, we expect developments in agent technology which will assist in the set-up and management of future services. In the opening of this section, we characterised a telepresence service as a reconciling of customer need for remote presence with appropriate media and control. As the options multiply, there will be increasing need for transparent automation in this matching of needs with available technologies. Current conferencing protocols can exchange details of terminal ‘capabilities’, such as coding schemes and camera control. Future agent-based solutions will extend to the negotiation of scheduling and media selection, using personal profiles, history logs and diary information.

In addition to simplifying service set-up, it is also important to provide familiar default interfaces which mimic established services in their interaction with both the environment and the technology. As new technology becomes widely accepted, this requirement for conventional interface metaphors will diminish. For example, the computer ‘desktop’ is a comfortable and flexible spring-board, but does not extend efficiently into immersive virtual environments, which literally open new dimensions in interface design. Greenhalgh et al [10], and Sidhu and Bowman [15] give particular consideration to interaction in virtual environments.

Picard and Cosier also explore a new dimension of user interfaces, in their overview of ‘affective computing’ [16]. They highlight the role of emotional intelligence in enabling human decisions and communications, and look forward to a future in which computers can support and mediate such interactions. As with the promise of virtual environments, increased processing power and innovative hardware will progressively overcome practical limitations which have traditionally moulded interfaces to the strengths and weaknesses of computers, not those of their human users.

3.2 Service assessment

In parallel with advances in individual technologies and interface design, we must progress our ability to assess and monitor complex multimedia communications, both during development and in service. Once again, the telephone provides instructive pointers, with a wealth of understanding on objective measures and also subjective performance assessment.

Subjective tests have been used for many years as an effective means of evaluating the transmission quality of telephony networks. In a typical test, subjects are asked to rate intelligibility of dialogue under a carefully controlled

set of network and environmental conditions, e.g. different levels of loss and noise. However, for live network testing there are both practical and cost advantages in using test equipment to provide objective assessment of systems. Hollier and Voelcker have developed such an objective system for audio quality assessment and their paper [17] describes an extension of their work to multi-modal assessment, for services which involve more than one sense. Multi-modal assessment presents a major challenge in understanding the psychological factors of importance and mapping these on to appropriate measurement technologies, but it is essential if we are to be able to objectively assess the perceived quality of future telepresence services.

3.3 Vertical markets

As we resolve both technical and service issues, the future of telecommunications will see increasingly flexible and ubiquitous multimedia telepresence progressively displacing ‘bread’n’butter telephony’. However, the commercial starting point in the face of such complexity must be a selection of ‘vertical markets’ with clear telepresence needs and quantifiable short-term benefits. These highly focused applications must be able to exploit the capabilities of early generations of specialised hardware, such as wearable computers, and we will also see a growing role for kiosks and other forms of public computing in providing wider access.

A number of papers point to early opportunities for telepresence:

- multi-player gaming is currently the leading market for residential multimedia ‘conferencing’, perceived demand for these services driving developments in both hardware and software, which aim to deliver compelling desktop virtual environments over relatively modest bandwidths — Powers et al [18] address long-term challenges in the scalability and performance of such applications, motivated initially by the latency and persistence requirements of games, but of equal relevance to other inhabited virtual environments,
- tele-medicine offers immediate benefits in cost and quality of care, and sets relatively clear goals for telepresence services — Garner et al [19] report on a range of early trials which stress the importance of focused design and integration of the overall service,
- tele-education is another important market, where telepresence can have an immediate impact — Fowler and Mayes [20] present a vision of learning as a social and collaborative process, ripe for the application of telepresence principles, reporting practical lessons from a range of robust prototypes which have been used in real educational settings,

- the world of finance is our final example of a vertical market for telepresence services — Heatley et al [21] describe an integrated desktop environment, which provides a tailored mix of information and communications services to dealers; the skills and interactions supported are typical of many industries in an information economy, and similar developments can be envisaged for other sectors.

3.4 Immersive conferencing

Finally, and looking further forward into the future, we outline a broader vision of two more generic telepresence scenarios with potentially widespread application. Our first vision is of 'immersive conferencing', a telepresence conferencing service which is 'better than being there'. Figure 2 illustrates the concept for two people seated at a desk. Large video images extend the desk 'through the screen' to participants at remote locations and a consistent, spatial audio environment gives a true sense of co-presence. Opportunities to go beyond a real life meeting arise through the use of intelligent processing in the interface, perhaps to provide access to on-line information services, an automatic secretary or even language translation.

We can conceive of a personal immersive conferencing workstation. In this 'SmartSpace' concept demonstrator, described by Traill et al [13], a wrap-around visual display and personal audio 'sound bubble' provide the sense of immersion, while the tablet is a flexible local desktop.

Figure 3 extends the concept to a purpose-built room. Wearable computing is used to identify participants, record and customise their interactions and track their location. A virtual 'greeting agent' handles security and access to the main conferencing area, where tracking information can be used to optimise both the hands-free audio and video environments.

3.5 Persistent communities

'Persistent communities' is a telepresence service which extends beyond real-time conferencing to meet the ongoing communications needs of a geographically distributed customer community. We referred earlier to the vision of 'Inhabited TV', a service in which professional broadcast material is mixed with social chat, and communities form around content and celebrities. We can envisage communities forming around a sport's team, soap opera or a pastime such as gardening. User involvement and participation is essential in building communities where

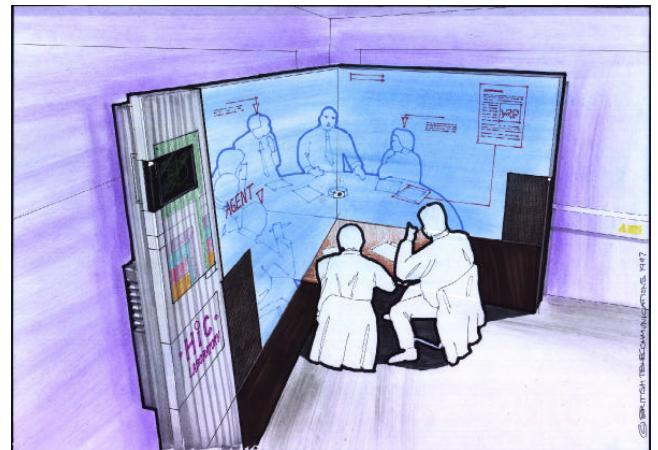


Fig 2 Desktop immersive conferencing.

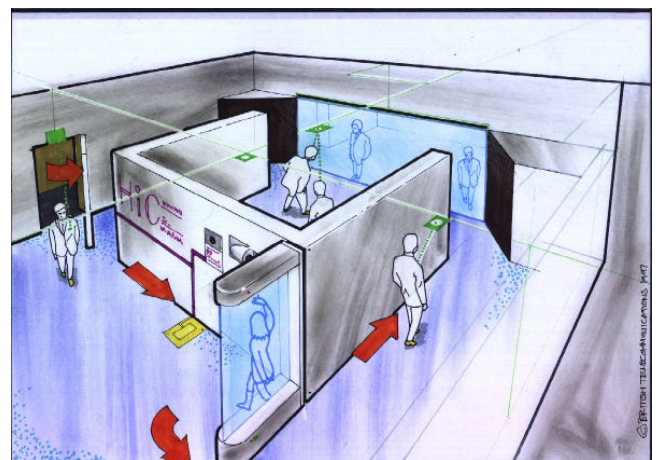


Fig 3 Room-based immersive conferencing.

interactive communications complements traditional passive broadcasting.

Another expression of this vision is the company shared space, which supports a community built around a brand or product. Such a service, or 'corporate portal', would provide elements of a high-street shop, a telephone support line, physical brochures and on-line information.

Portal communities would aim to personalise marketing and build customer relationships analogous to the proliferation of physical world club and loyalty schemes. They would also exploit the power of 'referral marketing' by promoting direct contact between existing and potential customers. Finally, they would target the cost saving achievable through 'self help' of the type currently practised on electronic news-groups and bulletin boards.

Figure 4 illustrates 'Going Spaces', a travel community, where members can experience possible destinations in the company of both experts and fellow travellers.

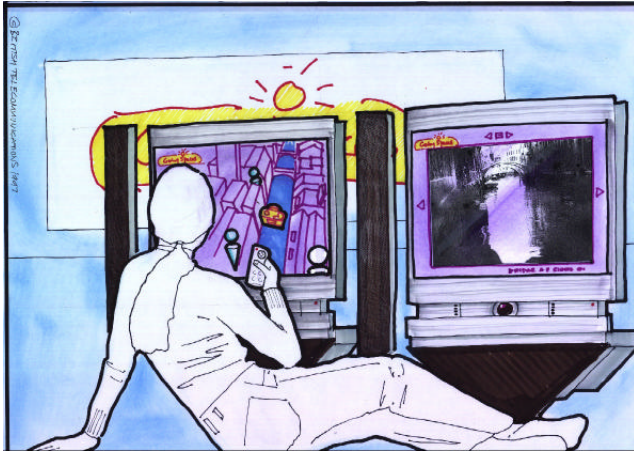


Fig 4 'Going Spaces' travel persistent community.

4. Conclusions

Multimedia telepresence is the future of telephony — diversity of media, matched to customers' needs and capabilities in delivering an appropriate degree of remote presence. However, we must not be blinkered in merely replacing and updating current communications channels. Our closing scenarios hinted at opportunities for telepresence to revolutionise existing processes. Much of the success of telephony has been achieved by meeting previously unrecognised needs, helping to solve problems that people did not know they had. Similar imagination and flexibility will see the emergence of novel telepresence services, which will again transform commercial and social life.

In outlining our vision we have highlighted a range of opportunities and challenges, many of which will be revisited in greater depth in the following articles. The telephone has set high standards for telepresence services, and an ongoing depth and breadth of research will be essential in meeting customer expectations.

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